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Deep Vision Project Report

Comparing CNN, ViT and Hybrid Architecture for Pneumonia Detection with Grad CAM Analysis

by

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Deep Vision Project Report Summary

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Abstract:

One of the key lung infection diseases is Pneumonia. To detect this lung disease, radiologists normally use chest X-ray images, but it takes a longer time. Also, expert knowledge, such as that of radiologists, is not always available. Deep vision models can help to detect pneumonia automatically by classifying X-ray images as normal or pneumonia, but how do we define which architecture works best, and how can we trust what the model has learned?

This study investigates and compares three vision architectures to identify pneumonia on a chest X-ray dataset of 5,856 images. Here, ResNet-50, ViT-Base (Vision Transformer), and TinyViT (a hybrid model that combines CNN feature extraction with transformer attention) are used. All three models were trained using 5-fold cross-validation with soft ensemble, making the results robust and directly comparable to the previous study.

Further in this study, we apply Gradient-weighted Class Activation Mapping (Grad-CAM and Grad-CAM++) to each model to visualize which regions of the given X-ray image the model focuses on when making its classification decision. Here, the Grad-CAM analysis results are shown in both qualitative and quantitative ways. The whole project provides practical guidance on architecture selection for medical image classification tasks.

Keywords: pneumonia detection, chest X-ray, CNN, Vision Transformer, TinyViT, Grad-CAM, explainable AI, medical image classification

1 Introduction

1.1 Background and Motivation

Millions of people suffer from lung disease around the world and many children and elderly people die because of lung disease. Pneumonia is one of the most dangerous lungs diseases and it is mostly diagnosed by using chest X ray images.However to read this kind of X ray images needs years of expertise radiologigirs who are not always avilable in the hospitals speacially areas with low medical resources.This motivates automatic diagnosis systems that can anlysze X rays automotically and support doctors to make faster decisions.

Rising of Deep vision models can supports radiologists by classifying these chest images automatically as normal or peumonia.Past studies have shown that two main families of deep learning architechtures have majorly impacted in image classification. **Convolutional Neural Networks (CNNs)** are designed to scan images using small kernel filters and they are really good at detecting local features of the given image. **Vision Transformers (ViTs)** are running a fundementally different apporach.They devide the given image into samll patches. It uses mechanism called *self-attention* to model relationships between every pair of patches. This allows ViTs to capture global feauture of the image rather than just local features.Recently, **hybrid architectures** combine convolutionals for efficient local feature extraction in early layers and attention for global understanding in later layers.This project majorly provide comparison of all three models under identical conditions and measure the quality and the reliability of the visual reasoning.

Beyond accuracy, there is a second critical question: *can we trust what the model has learned?* A model that achieves high accuracy by focusing on irrelevant parts of the image (such as the image border or a hospital equipment marker) is not safe for clinical use. **Gradient-weighted Class Activation Mapping (Grad-CAM)** is a widely used technique that produces a heat map showing exactly which regions of the image the model focused on when making its decision. Phase 2 of this project applies Grad-CAM systematically across all three architectures to answer this question.

1.2 Problem Statement

This Project discussed,three important questions remain open:

- Most studies compare CNNs *or* ViTs, not both together.
- This projects base report results from a single training run, which can be unreliable because deep learning is sensitive to random initialisation.
- Most studies apply explainability tools (like Grad-CAM) to only one model, rather than comparing how different architectures visualise their reasoning.

This project focuses above gapes by training CNN (ResNet-50), a Vision Transformer (ViT-Base/16), and hybrid model (TinyViT-21M) under same controlled setup.Then apply Grad-CAM and Grad-CAM++ to all three in Phase 2.

1.3 Contributions

Project Contributes

1. **Adding a hybrid architecture.** TinyVit-21M is a true CNN-Transformer hybrid which introduces alongside the pure CNN and pure Transformer baselines.The The base paper by Alqutayfi et al. (2025), which this project extends, did not include any hybrid model.
2. **A reliable evaluation protocol.** Uses 5 fold cross validation with soft ensemble instead of a single run.
3. **A fair, controlled comparison.** All three architectures use the same training set up and evaluation metrics. Only the learning rate differs, and for a good reason (explained in Chapter 3).
4. **Systematic explainability (Phase 2).** Grad-CAM and Grad-CAM++ will be applied to all three architectures, with qualitative and qunatitative anlysis.

2 Literature Review and Research Gap

2.1 ResNet-50: The Pure CNN Baseline

ResNet-50 [1] is a convolutional neural network whose speciality is the *residual connections* which is a shortcut path that lets the training signal skip over layers and make very deep networks trainable.ResNet-50 contains about 24 million parameters and it is widely trusted baseline for medical domain image classification tasks.Bacuse CNN analyze images through local kernel filters, which they naturally focus on local patterns. This suits well for chest X ray images where pneumonia appears localized cloudy regions.

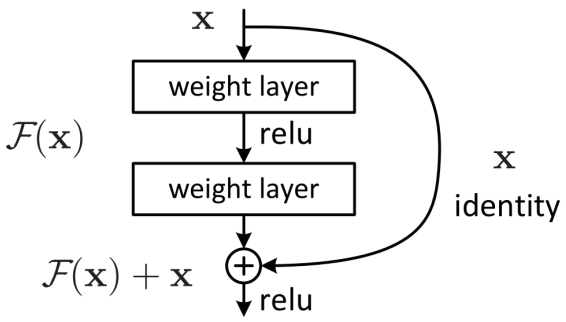


Figure 1: The residual building block used in ResNet.[1]

2.2 ViT-Base/16: The Pure Vision Transformer

The Vision Transformer [2] applies the transformer architechture which is originally designed for language to images.It seperates an image into non overlapping 16×16 pathces, treats each splitted patch like a word, and uses self attention mechanism to model relationships between all patches. This ViT base version has about 86 million parameters.Unlike CNNs, ViTs do not have built in assumption that nearby pixels are related,so they made this architechture to model the whole image globally but they need large training datasets which is a know draback on small medical datasets.

2.3 TinyViT-21M: The Hybrid Architecture

TinyViT-21M [3] combines CNN and Tranformer ideas in one model .Tiny ViTs early stages uses convolutional blocks (MBConv blocks,the efficient convolutions from EfficientNet) to capture the local details of given image and in its later stages it uses self attention blocks to capture global relationships. This architure first processes the image like a CNN and then like a Transformer.It has fewer parameters (21 millions) than both ResNet-50 and ViT Base, and designed to both accurate and efficient.This local and then global processing is why TinyViT is the hybrid candidate in this study.

2.4 Research Gaps

The base paper [4] compared several CNN and ViT models on pneumonia chest X-rays images, and used Grad-CAM to visualise model decisions [4]. It is a valuable reference for this project, but it left two specific gaps:

Three Gaps This Project Addresses

1. **No hybrid architecture.** The base paper compared CNNs and ViTs but did not include any hybrid model.
2. **Grad-CAM only on the best model.** The base paper applied Grad-CAM only to the single best-performing model.They left Grad CAM++ anlysis as future work.

3 Methodology

3.1 Overview: The Phase 1 Pipeline

Figure 2 shows the full pipeline. This pipeline applied to each of three architechtures exactly through a same setup, so any differences in the final resuls is due to the architechture itself not the depend on the training set up.

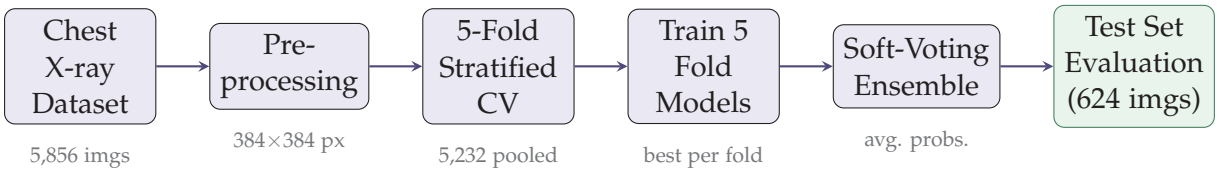


Figure 2: The Phase 1 pipeline applied identically to ResNet-50, ViT-Base/16, and TinyViT.

3.2 Dataset

This project uses the **Kermany Chest X-ray Dataset** [5], the same dataset used in the base paper.It contains chest X-ray images labeled as either NORMAL (healthy lungs) or PNEUMONIA (infected lungs),divided into 5216 training images, 16 validation images, and 624 test images. The images are grayscale and the training data is imbalanced, with roughly three times as many pneumonia images as normal images.

Table 1 shows how the dataset is divided.

Table 1: Kermany chest X-ray dataset composition.

Split	Normal	Pneumonia	Total
Training set	1,341	3,875	5,216
Validation set (original)	8	8	16
Pooled for CV (train + val)	**1,349**	**3,883**	**5,232**
Test set (held out)	234	390	624
Total	1,583	4,273	5,856

The original paper used validation set as only 16 images (8 normal and 8 pneumonia).This is too small to give a reliable measurement of how well a model is performing.If the model happens to get 5 of those 8 pneumonia case right or wrong by chance, the validation score changes dramatically.

Instead of that here, training and validations sets are combine together (5,232 images total) then divided into 5 equal folds for cross validation.This way each validation fold contains about 1,047 images.It performes a much more reliable sample while the 624 image test set stays completely held out as the final performance measure.

3.3 Preprocessing

First we have to use same preprocessing steps before they go into the any kind model.Every image is resized to **384×384 pixels**, which matches the resolution used in the base paper. It also ensures that all three architectures receive the same image size.

X-ray images are by default **grayscale** (one channel of pixel values). But all three models were pretrained on ImageNet, which contains color (RGB, three-channel) photographs. Because of this reason model expect a three channel input.To do this, we simply **copy the grayscale channel three times** to create a three-channel image.Now the models can process the images directly while preserving the pretrained weights of ImageNet. Here we use ImageNet mean and standard deviation ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) to normlaise the pixel values. This makes the input values comparable to what the models saw during ImageNet pre-training and helps them converge faster.During training, random augmentations are applied to the images to increase the diversity of what the models see.This also reduces the chance of overfitting. During validation and testing, no augmentations are applied, only resizing and normalisation.

3.4 The Three Architectures

All three architecures were already trained on a large dataset (ImageNet , with 1.2 million photos) , then we use to train on smaller dataset (chest X rays).Models know how to detect features like edges, textures and shapes; fine tuning teaches it to apply those skills to the specific task of idetifying pneumonia.This approach works much better than training from random weights when the medical dataset is small, which is the case here.

Table 2: The three architectures fine-tuned in this project.

Architecture	Type	Params	Learning Rate
ResNet-50	Pure CNN	≈24M	1×10^{-3}
ViT-Base/16	Pure Transformer	≈86M	1×10^{-4}
TinyViT-21M	CNN+Transformer hybrid	≈21M	1×10^{-4}

Why Different Learning Rates?

Learning rate controls how much the model updates its weights after each training step. ResNet-50 uses 1×10^{-3}, which is the standard CNN learning rate.ViT-Base/16 and TinyViT-21M use 10 times smaller learning rate 1×10^{-4}. In early experiments ViT-Base was trained at 1×10^{-3} , and the training become unstable (loss spiked and did not converged).Theh base paper used same learning rate for all models which likely to reduce the perfomance of ViT results. Using the correct architecture with suitable learning rate is one of the key improvements in this project.

3.5 Handling Class Imbalance

Here our dataset contains more pneumonia images than normal images.The model can be biased to predicting pneumonia.A class weighted loss fucntion can be solve this problem.Here we set the weight for each class proportional to the inverse of how often it appears in training data. With this we can give more improtance to the rare class (Normal).

These weights are computed separately for each fold from that fold's training data, because the class counts vary slightly between folds.These weights are then applied inside the weighted cross-entropy loss, where each image's loss is multiplied by the weight of its true class:

$$\mathcal{L} = -\left[\, w_{\text{Normal}}\, y_{\text{Normal}} \log(p_{\text{Normal}}) + w_{\text{Pneumonia}}\, y_{\text{Pneumonia}} \log(p_{\text{Pneumonia}}) \,\right]$$

3.6 Training Setup

Below table shows the training hyperparameters that used for all three architechtures. The same configuration was applied to every model, so that any difference in results can be attributed to the architecture itself rather than the training procedure.

Table 3: Training hyperparameters used for all three architectures.

Setting	Value
Optimiser	AdamW (weight decay $= 1 \times 10^{-2}$)
Learning rate	1×10^{-3} for ResNet-50; 1×10^{-4} for ViT-Base/16 and TinyViT-21M
Loss function	Class-weighted Cross-Entropy
Batch size	16 images per step
Epochs per fold	10
Checkpoint criterion	Best validation F1-Score (saved per fold)

3.7 Evaluation Metrics

We fllow the same matrics as the base paper: accuracy, precision, recall, and the F1 score. Our primary metric is weighted F1 score bacause it balances precision and recall without affecting the class imbalance scenario.

Additionally we reported below metrics as well.

- **Recall (Sensitivity)** - how many true pneumonia cases are correctly detected. This is the most important metric clinically, since missing a sick patient is dangerous.
- **Recall (Normal)** - how many true normal cases are correctly classified.
- **Cohen's Kappa** - how much better than random chance the model performs.
- **Confusion Matrix** - the full breakdown of true/false positives and negatives, showing exactly which errors the model makes.

3.8 Methodology for GradCam Analysis

In phase 1 we measure how different architechtures classify X rays accurately.In phase 2 we find *why did the model make that decision?*. Gradient weighted Class Activation Mapping [6] is a method to visualise which sections of an image used to make its decision.It reads newtowrk's own internal computations instead of guessing.

The major difference between GradCAM and GradCAM++ [7] is in step 3.GradCAM++ is highlights multiple distinct regions more sharply while GradCAM gives a single smoother highlight.

3.8.1 Selecting One Representative Model Per Architecture

GradCam only needs the gradient information from a single model. Therefore for this GraCAM anlysis one representative fold is seleceted for each architechture.Here we selected the fold as whose test set F1 score is closest to that architechture's average F1 score across all five folds.

Table 4: Representative fold selected for Phase 2 explainability analysis.

Architecture	Type	Fold Used	Fold Test F1
ResNet-50	CNN (pure)	Fold 4	0.928
ViT-Base/16	Transformer (pure)	Fold 2	0.908
TinyViT-21M	Hybrid (CNN+Attention)	Fold 0	0.943

To apply GradCAM we selected the final feature layer, which is standard choice. This layer catuptures most abstract, high level patterns. For all architechtures we use this selected layer so the comparison is fair.

Table 5: Target layer used for Grad-CAM in each architecture.

Architecture	Target Layer	Special Handling
ResNet-50	Last convolutional block	None needed
ViT-Base/16	Final transformer block norm	Token $\rightarrow$ grid reshape
TinyViT-21M	Last stage conv layer	None needed

Unlike CNN ViT does not produce a 2D spatial feature map.It outputs a sequence of tokens in one per image patch.Before apply the GradCAM heatmap these tokes should be rearrange back into 2D grid.

3.8.2 Evaluation metrics for Grad-CAM analysis

In out base paper they have used a radiologist evaluation to compare thier GraCAM heatmaps.But in this study we used heatmaps as qualitative analysis and two quantitative metircs for better evaluations in this phase 2.

1. **Inside-Body Fraction** - This metric finds *Where does the model look?* We want to ensure that the model is looking inside the patient's body or not.

2. **Deletion AUC** [8] - This metric finds *Is the heatmap actually driving the decision?* a faithful heatmap causes model confidence to drop quickly when the highlighted pixels are removed.

4 Phase 1 Results and Discussion

4.1 Classification Performance

Table 6 shows the final resutls for all three architechtures which is evaluated on the 624 test images.

Table 6: Ensemble test-set performance on the Kermany chest X-ray dataset (624 images). Recall denotes pneumonia sensitivity (recall on the pneumonia class).

Architecture	Acc.	F1	Recall	Kappa	Paper F1
ResNet-50	0.9359	0.9346	0.9949	0.8589	0.7601
ViT-Base/16	0.9279	0.9267	0.9821	0.8420	0.7887
TinyViT-21M	**0.9487**	**0.9480**	0.9949	**0.8879**	no hybrid

Key Findings: Classification Performance

- All three architechtures achieved strong results above the base paper values.
- **TinyViT-21M (hybrid) achieved the highest F1-Score (0.9480)** while using the fewest parameters (≈21M), outperforming both the pure CNN (ResNet-50, 0.9346) and the pure Transformer (ViT-Base/16, 0.9267).
- ResNet-50 ranked second (F1 = 0.9346), outperforming ViT-Base/16 (F1 = 0.9267), despite ViT-Base having nearly four times as many parameters (≈86M vs ≈24M).
- TinyViT-21M was also the strongest on every other metric, with the highest accuracy (0.9487), Cohen's Kappa (0.8879), and the fewest false alarms (30 of 234 normal cases), while missing only 2 of 390 pneumonia cases.

4.2 Cross-Validation Stability

One datasplit will not give us the reliable results.We recoded best validation F1 score of each model in five folds. Below low standard deviation scores show that every architechture learning happened consistently.

Table 7: Cross-validation stability of each architecture.

Architecture	Mean Validation F1	Std. Deviation
ResNet-50	0.9772	0.0076
ViT-Base/16	0.9453	0.0149
TinyViT-21M	**0.9790**	**0.0016**

Key observations:

- **TinyViT-21M is the most consistent** (std = 0.0016), followed by ResNet-50 (std = 0.0076).

- **ViT-Base/16 is the least stable** (std = 0.0149), proves the known sensitivity of pure Vision Tranformers in small datasets.
- TinyViT-21M has both the highest mean F1 and the lowest variance, the most accurate and the reliable architecture.

4.3 Confusion Matrices

Figure 3 shows the confusion matrix for each architecture's on the 624-image test set.

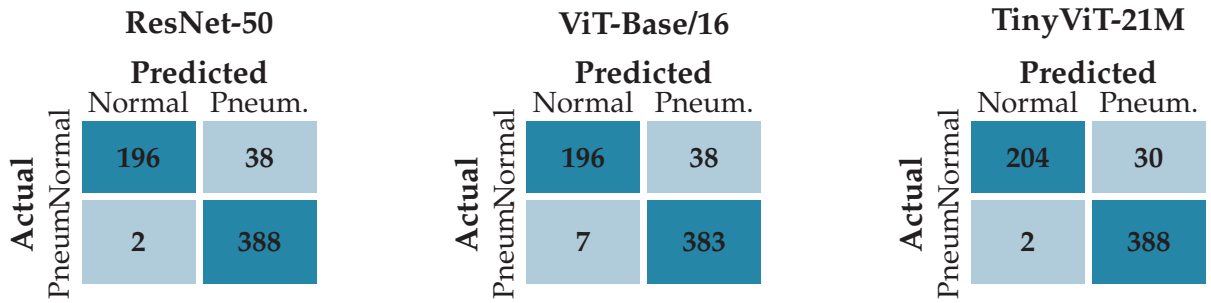


Figure 3: Confusion matrix for each architechture.

The off-diagonal cells reveal the two error types:

- **False negatives** (Pneumonia → Normal): a sick patient is missed.This is the most dangerous error.
- **False positives** (Normal → Pneumonia): a healthy patient is sent for unnecessary follow-up.

TinyViT-21M makes the fewest errors. **ResNet-50** has more false alarms (38) compared to Tiny ViT. **ViT-Base/16** misses the most pneumonia cases (7), matching its lower recall.

4.4 Effect of the Hybrid Architecture

Does Combining CNN and Transformer Help?

TinyViT-21M performed better than ResNet-50 and ViT-Base/16, while using fewer parameters. The likely reasons:

- **Local features:** the early convolutional stages detect local patterns (cloudiness, opaque patches) that indicate pneumonia.
- **Global context:** the later attention stages see how these patterns are distributed across the lung fields.
- **Less data needed:** the convolutional early layers give the model a head start, so it needs less labelled data to learn. This an advantage on a small medical dataset.

5 Phase 2 Results and Discussion

In phase 1 showed all three architechtures classify pneumonia with high accuracy.But high accuracy does not automaticcaly mean that the model is looking at the right place.Model can reach the correct answer with the wrong reason.This kind of failures we can say as a shortcut learning and it makes a model unrealible and unsafe for real clinical setting.

Phase 2 uses **Grad-CAM** and **Grad-CAM++** to answer three questions:

1. **Where does each model look?** Does the model's attention stay inside the patient's body, or does it leak onto the background or image borders?
2. **Is the heatmap trustworthy (faithful)?** Do the highlighted regions actually drive the model's prediction, or are they just a visually pretty pattern that does not correspond to real model behaviour?
3. **Does Grad-CAM++ improve on standard Grad-CAM?** The base paper (Alqutayfi et al., 2025) suggested Grad-CAM++ as future work but never applied it. This chapter reports whether it actually helps.

5.1 Qualitative Comparison

Figure 4 shows the Grad-CAM and Grad-CAM++ heatmaps for six representative X-rays.

Even from a visual inspection, three clear patterns emerge:

- **ResNet-50 (CNN):** The heatmaps are **compact and lung-focused**. For the pneumonia cases red regions concentrates direcly on the cloudy or opaque area in the lung. For normal cases the focus stays on the clear lung fields.Grad-CAM and Grad-CAM++ look almost identical for this architechture.
- **ViT-Base/16 (Transformer):** The heatmaps are **spread-out and diffuse**. Attention covers entirely chest cavity and drifts onto the image border or the orientation marker.
- **TinyViT-21M (Hybrid):** The heatmaps fall **between the other two**.This one is more clear than the ViT but wider than the pure CNN. Its Grad-CAM++ maps are noisier then its Grad-CAM maps.

Here heatmaps alsways highlight some region the model used to reach its decision itself. Not the location of the disease.A red region on a correctly classified normal case means, the model evaluate that area and found it clear.

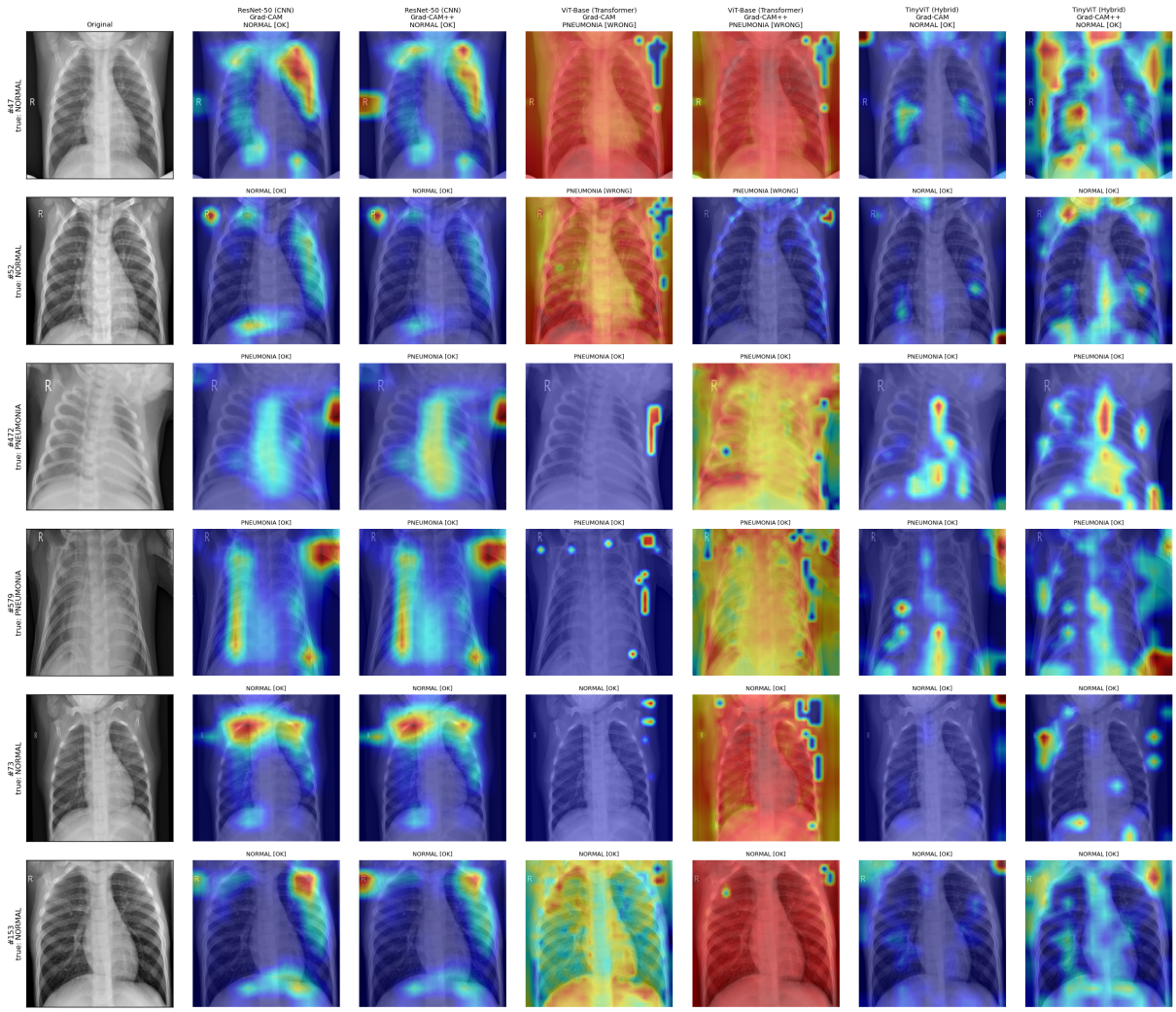


Figure 4: Qualitative heatmap comparison across all three architectures. Each pair of columns shows Grad-CAM (left column) and Grad-CAM++ (right column) for one architecture.

5.2 Quantitative Comparison

5.2.1 Inside Body Fraction

This metric measures what percentage of the total heatmap intensity falls inside the patient's body.When this is higher we can say our classication was done good.

Table 8: Inside-body fraction for each architecture and method.

Architecture	Grad-CAM	Grad-CAM++
ResNet-50 (CNN)	**0.925**	**0.928**
TinyViT-21M (Hybrid)	0.829	0.824
ViT-Base/16 (Transformer)	0.775	0.752

The ranking is: **CNN** > **Hybrid** > **Transformer**.

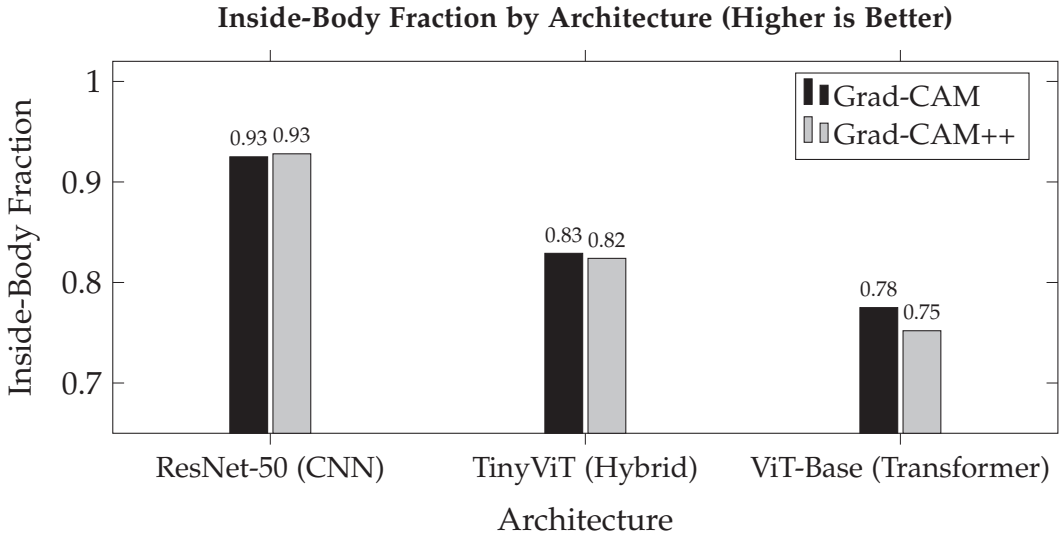


Figure 5: Inside-body fraction for all three architectures. The CNN (ResNet-50) keeps the most attention inside the patient's body (≈93%); the pure transformer (ViT-Base) leaks the most (≈78%).

- **ResNet-50** keeps ≈93% of its attention inside the body.
- **TinyViT-21M** keeps ≈83% , its convolutional early stages introduce a natural spatial preference that keeps attention more focused.
- **ViT-Base/16** keeps only ≈78%, meaning roughly 1 in 5 units of attention is on irrelevant regions outside the patient's body.

5.2.2 Deletion AUC

A heatmap can look realistic but misleading. The delection AUC find this by removing the highlighlited pixels from the image should cause the model's confidence to drop quickly, if the heatmap genuinly marking the important regions.

Table 9: Deletion AUC for each architecture and method.

Architecture	Grad-CAM	Grad-CAM++
ResNet-50 (CNN)	**0.532**	**0.548**
TinyViT-21M (Hybrid)	0.621	0.649
ViT-Base/16 (Transformer)	0.794	0.768

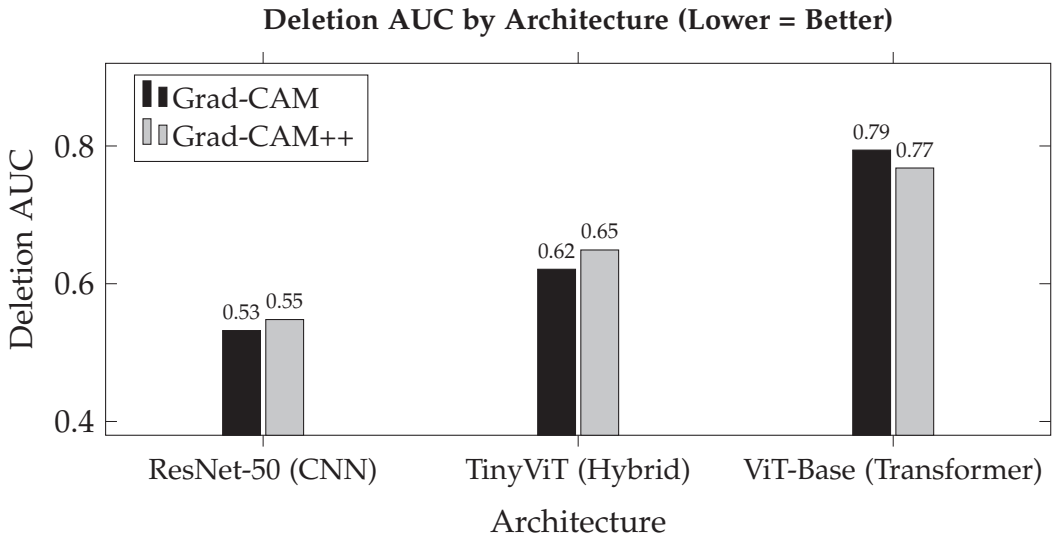


Figure 6: Deletion AUC for all three architectures. A lower bar means the heatmap correctly identified the important regions. ResNet-50 (CNN) is the most faithful; ViT-Base (Transformer) is the least faithful.

The ranking is again **CNN** $>$ **Hybrid** $>$ **Transformer**, identical to the localization result.

- **ResNet-50** has the lowest deletion AUC, meaning removing its highlighted pixels causes the model's confidence to drop the quickly.Its heatmap is the most faithful.
- **TinyViT-21M** is intermediate (0.621).
- **ViT-Base/16** has the highest deletion AUC (0.794), meaning its heatmaps are the least faithful. This matches the qualitative results of ViTs.

The base paper left Grad-CAM++ as future work.Also Grad-CAM++ is not cocnsistently outperform standard Grad-CAM on this classification task.

Comparison of Grad-CAM and Grad-CAM++

- **ResNet-50:** Grad-CAM++ marginally worse on faithfulness (deletion AUC 0.532 $\rightarrow$ 0.548); nearly the same on localization.
- **TinyViT:** Grad-CAM++ slightly worse on both metrics; visually noisier.
- **ViT-Base:** Grad-CAM++ slightly better on faithfulness (0.794 $\rightarrow$ 0.768) but slightly worse on localization (0.775 $\rightarrow$ 0.752).

All the differeces are small ($\leq$0.03). Grad-CAM++ does not give clear advantage for this classification task with any of the three architechtures tested.

5.3 Discussion of final Results

Both metrics produce same rankings.**ResNet-50 (CNN)** > **TinyViT (Hybrid)** > **ViT-Base/16 (Transformer)** and the qualitative heatmaps were agreed, so this ranking is well supported.

Reasons for the Rankings

The architecture explains the result directly:

- **ResNet-50 (CNN):** Convolutional kernels scan local pathces of the image.This helps the network a strong spatial prior and it naturally focuses on compace, localized regions.
- **ViT-Base/16 (Transformer):** Here evey image patch attends to every other patch. There is no spacial locality bias.Because of that heatmaps are diffused as a direct drawback.
- **TinyViT-21M (Hybrid):** Here early convolutional layers intorduce a local spacial bias like ResNet. Then the later attention layers capture global context like ViT.This architechtures results sit between the two in both metrics.

TinyViT gives the best classification accuracy, ResNet-50 gives more trustworthy and spacially precise explonations. The decison between them depends on whether interpretability or raw accuracy matters more in a given application.

6 Conclusion

In summary, the hybrid TinyViT-21M is the best choice for this task, combining the highest accuracy, the most stable cross-validation results, and the fewest parameters. Phase 1 confirms that combining convolution and attention is effective for chest X-ray pneumonia detection, and provides a reliable foundation for the explainability analysis in Phase 2.

In phase 2 generates heatmaps from all three architechtures on two independent metrics (inside body fraction and deletion AUC). ResNet-50 was the msot interpretable, TinyViT intermediate and ViT base the most diffuse.Grad-CAM++ gave no consistent gain over Grad-CAM. Finally we found TinyViT is the best classifier while ResNet-50 is the most trust worthy to explain, making it safer and reliable choice for clinical use cases.

6.1 Limitations

- **Binary classification only.** The models only separate Normal from Pneumonia, not bacterial from viral pneumonia or other lung conditions.
- **Single dataset.** All data comes from one hospital, so performance may differ elsewhere.
- **No speed or memory measurement.** Only accuracy was measured, not inference time or memory use, which also matter for deployment.

7 Declaration on the Use of AI Tools

AI-based tools were used during this project to assist with parts of the coding and the preparation of this Latex report, mainly for code suggestions, debugging, language refinement, and LaTeX formatting. All AI-generated content was reviewed and verified by the author, and the ideas, analysis, and conclusions presented are my own.

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